

Using Brain Imaging Data Structure (BIDS) for reproducible research

22nd july 2025

TU Dresden

Sein Jeung & Julius Welzel

How it all started



Sein Jeung

Technische Universität Berlin

Interest in extending Brain Imaging Data Structure (BIDS) spec to motion data?

Question Asked December 2, 2019

Dear all, motivated by a variety of MoBI studies currently being conducted in our group, I am thinking of proposing a new Brain Imaging Data Structure (BIDS) extension that addresses MoBI-specific needs for data sharing. This would also encompass flatscreen navigation data or 2D position streams from more conventional studies, so we can easily compare between recordings ... [Read more](#)

Answer this question

Comment | [Open access](#) | Published: 02 July 2024

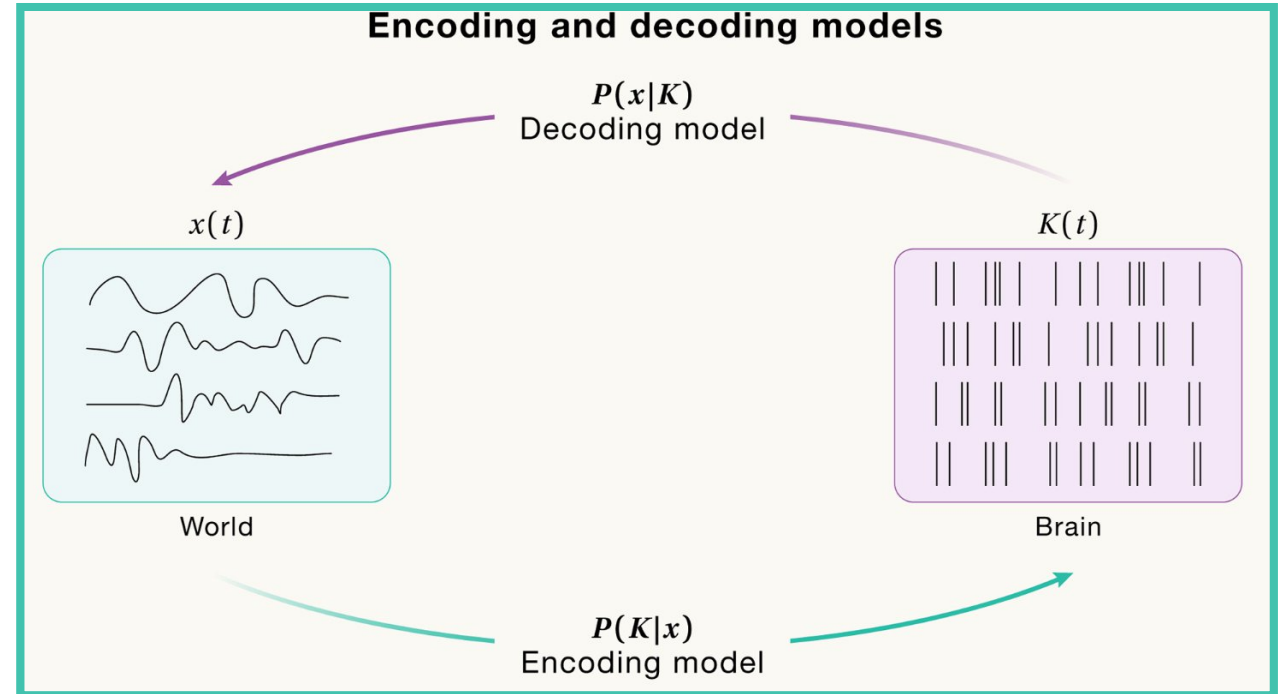
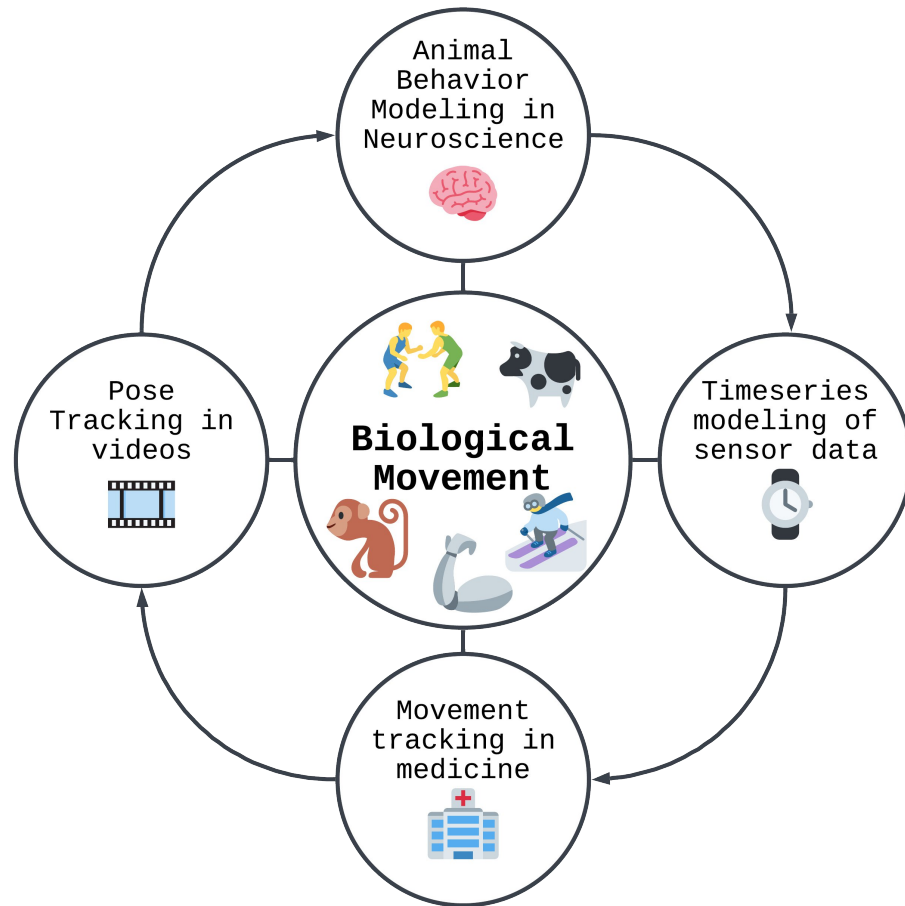
Motion-BIDS: an extension to the brain imaging data structure to organize motion data for reproducible research

[Sein Jeung](#) , [Helena Cockx](#), [Stefan Appelhoff](#), [Timotheus Berg](#), [Klaus Gramann](#), [Sören Grothkopp](#), [Elke Warmerdam](#), [Clint Hansen](#), [Robert Oostenveld](#), [BIDS Maintainers](#) & [Julius Welzel](#) 

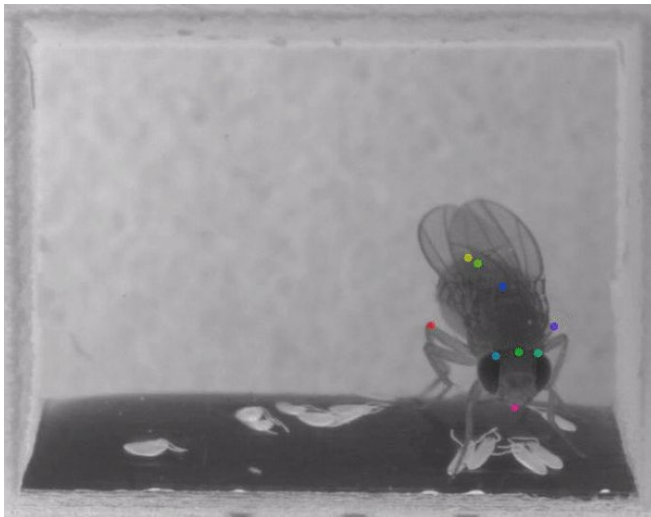
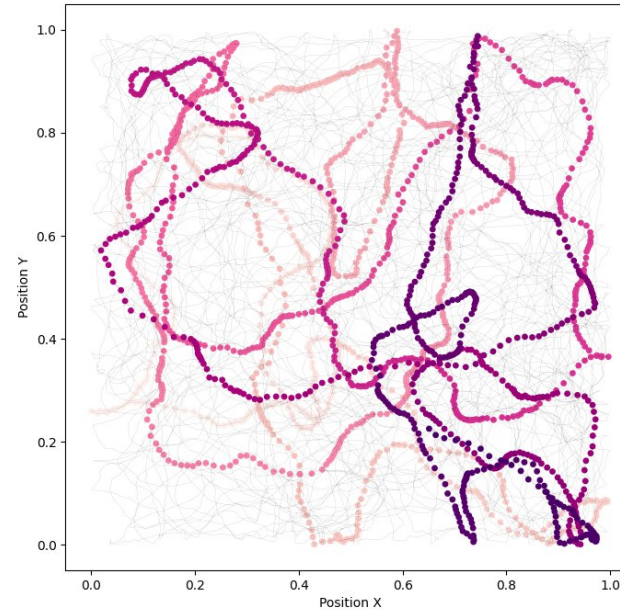
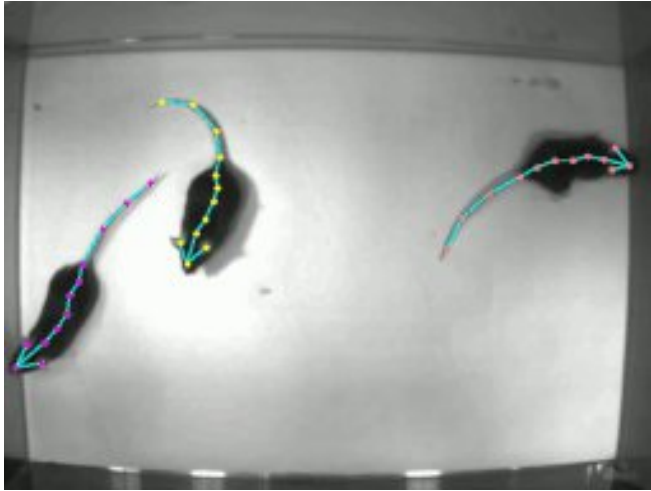
[Scientific Data](#) **11**, Article number: 716 (2024) | [Cite this article](#)

 [Save article](#)

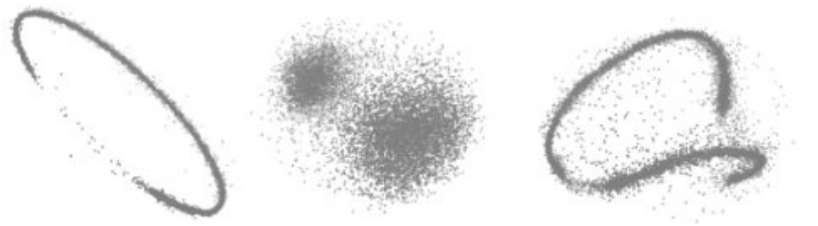
7285 Accesses | **26** Citations | **8** Altmetric | [Metrics](#)



What we can learn from animal research



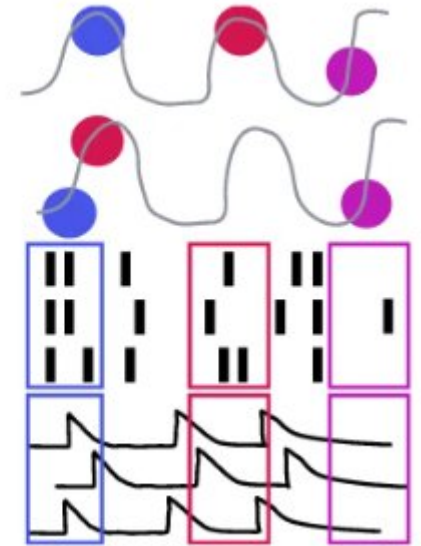
Position only Direction only Position + Direction



Behaviour labels

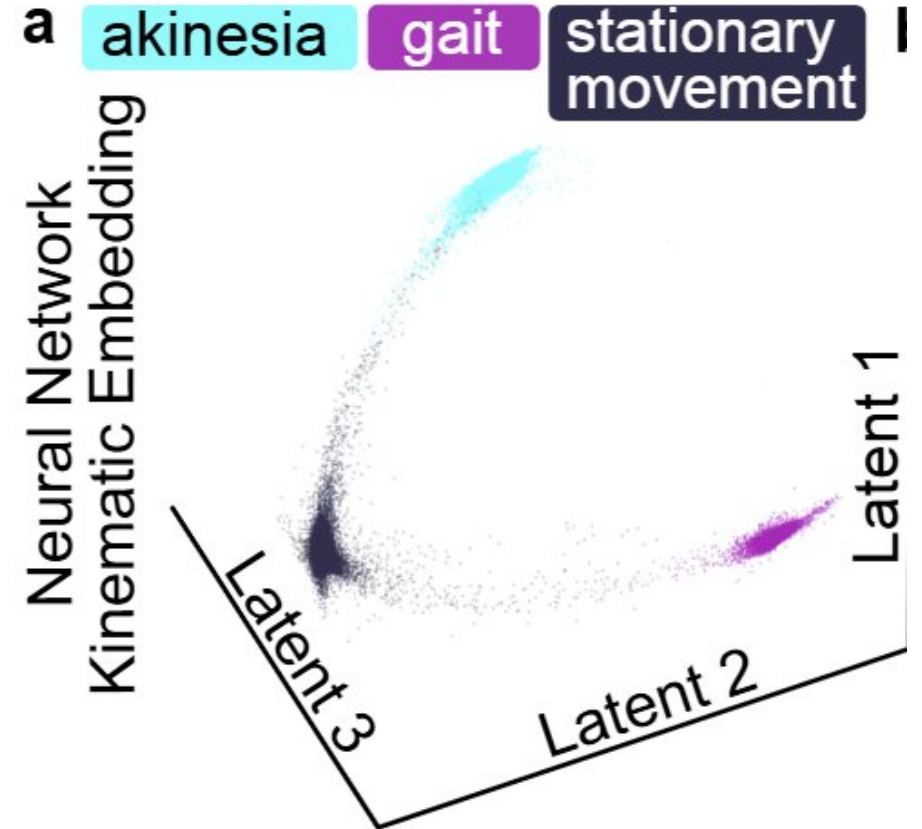
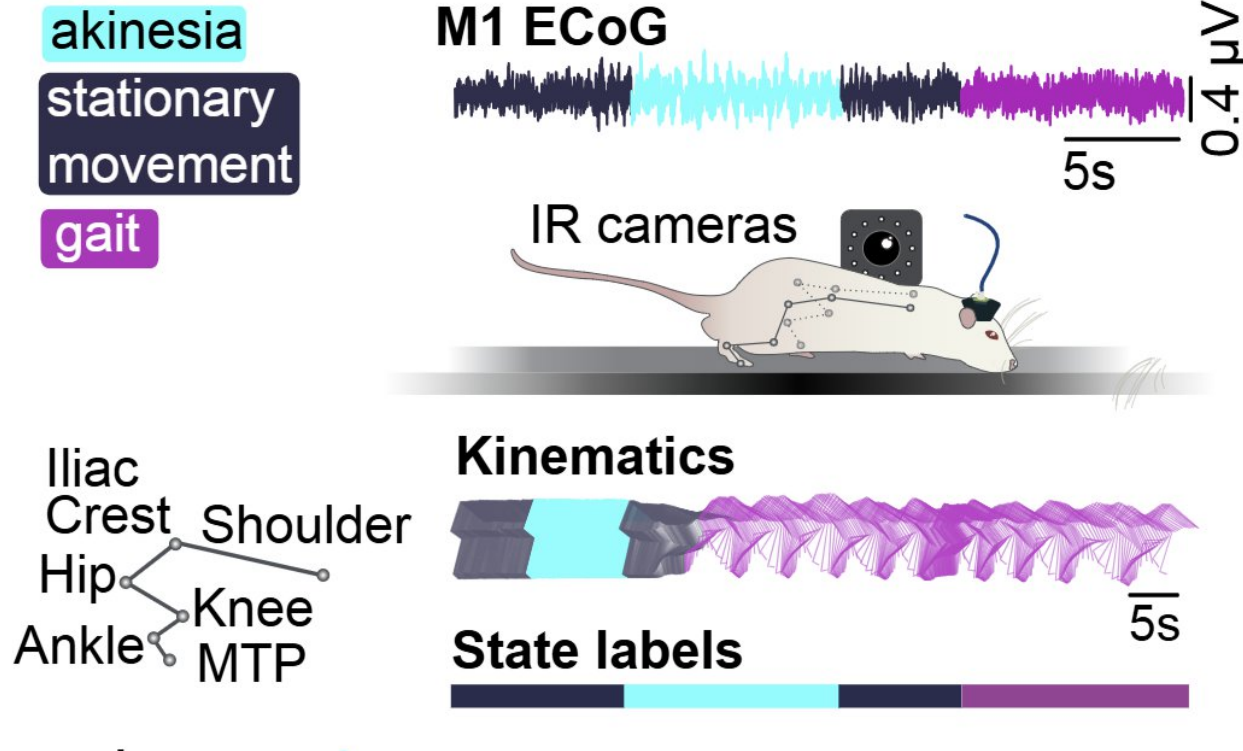
Time labels

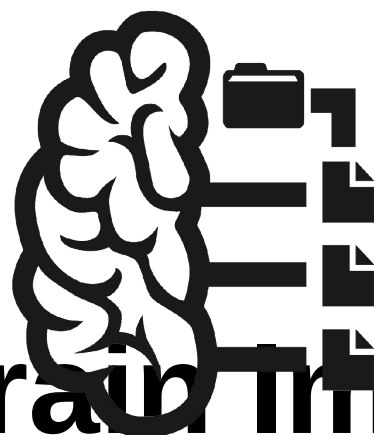
Neural data (N)



Neurobehavioral “phenotyping”

a





BIDS

**The Brain Imaging Data Structure
(BIDS)**

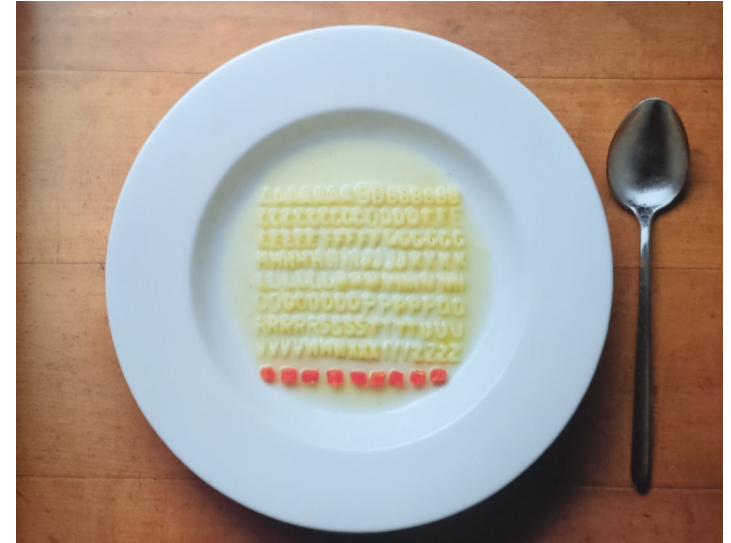
BRAIN IMAGING DATA STRUCTURE

What is BIDS?



Dataset info

Motion



Aims of BIDS



The Structure (Container)

- **Hierarchy:** Standardized Subject → [Session] → modality folders.
- **Predictable Naming:** Key-value pairs allow automated parsing.
- **Simple Formats:** Uses non-proprietary standards (.edf, .tsv).

Metadata (Context)

- **JSON Sidecars:** Every data file is paired with a .json file containing critical metadata details (e.g. sampling rate).
- **Machine-Readable:** Allows software (fMRIPrep, fieldtrip) to easily read data.

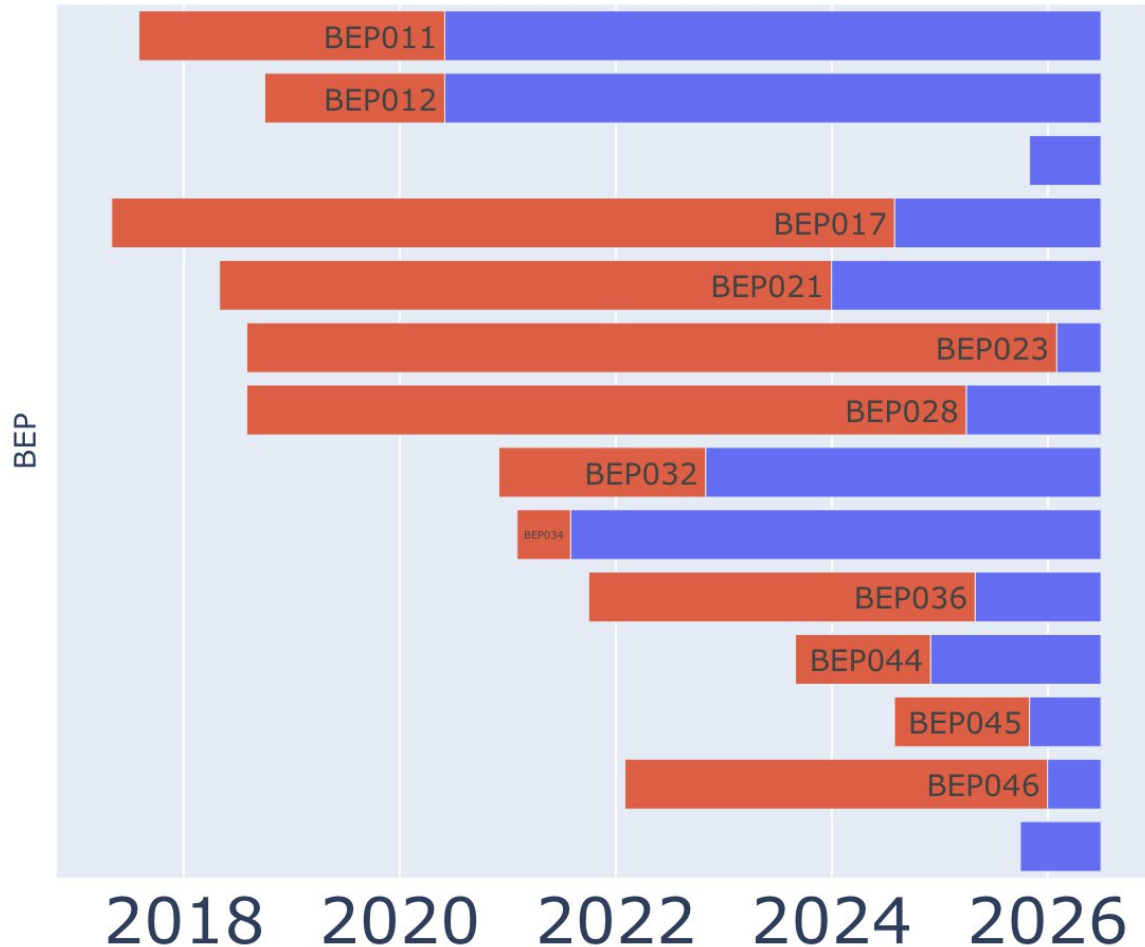
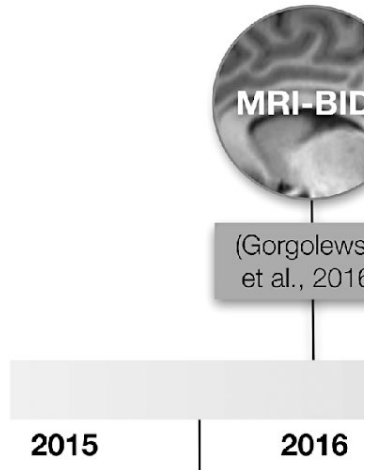
- **Beyond “Brain” Imaging:** dataset, participant, events

<https://bids-specification.readthedocs.io/en/stable/>



The Past, Present, and Future of BIDS

BIDS history



Resource

- Pull Request
- Google Doc



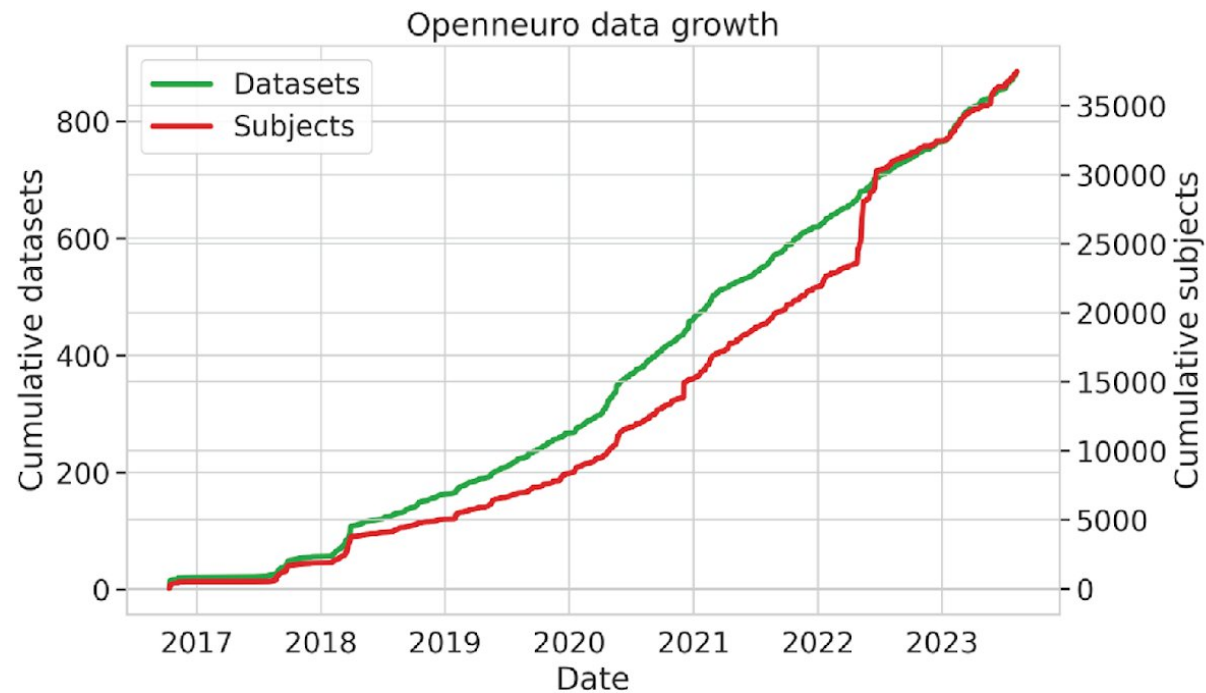
Luke et al.,
2023)



(Jeung et
al. 2023)

2023

The Past, Present, and Future of BIDS



 **OpenNEURO**

SEARCH SUPPORT DOCUMENTATION

OpenNeuro EEG

OpenNeuro added support for EEG datasets in 2019 when EEG was incorporated into the BIDS standard.

7.926 Participants **200** Public Datasets

Search EEG Portal

[Search at the participant-level with Neurobagel ?](#)

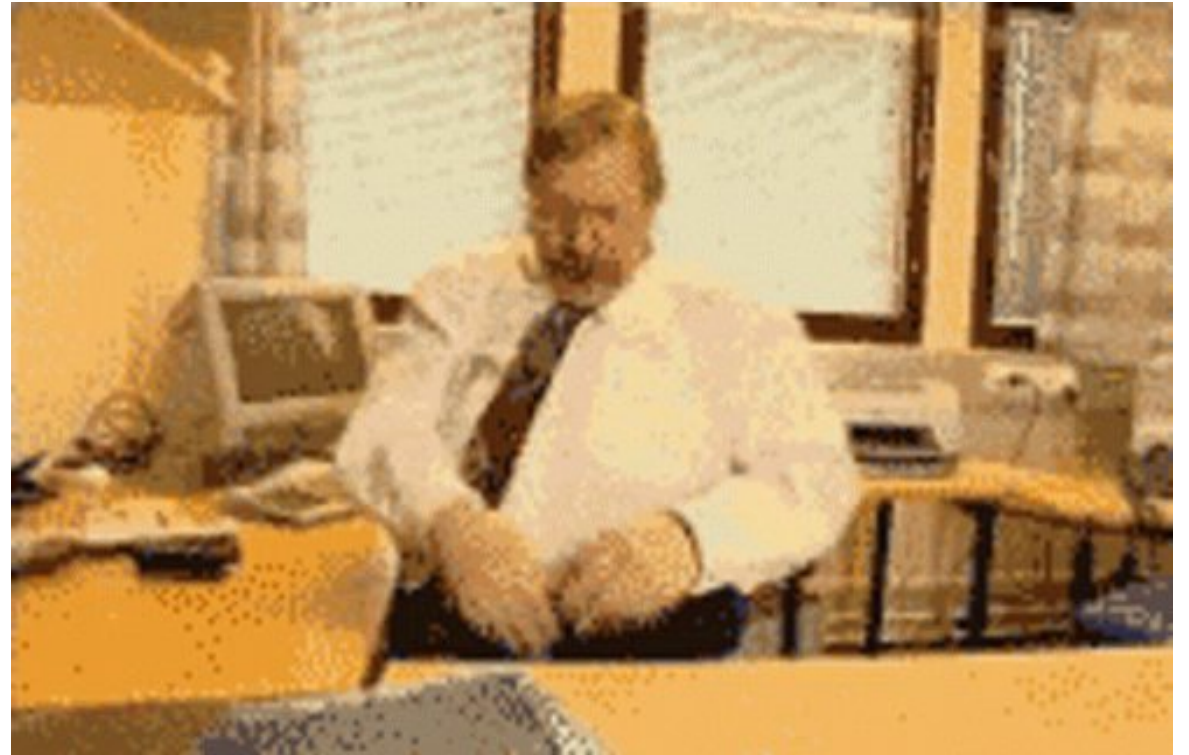
Keywords ?

These filters return **70** datasets:

KEYWORD:

MODALITY:

My favourite two BIDS principles



Requirement levels

Level of requirement **for** files and **within** files:

1. **REQUIRED**

2. **RECOMMENDED**

3. **OPTIONAL**

```
├─ README.md
├─
├─ dataset_description.json
├─ participants.tsv
├─ sub-001
│   └─ ses-01
│       ├── eeg/
│       └─ motion/
├─ ses-02
└─ sub-002
```

participant_id	age	group
sub-001	34	C
sub-002	12	P
sub-003	33	M

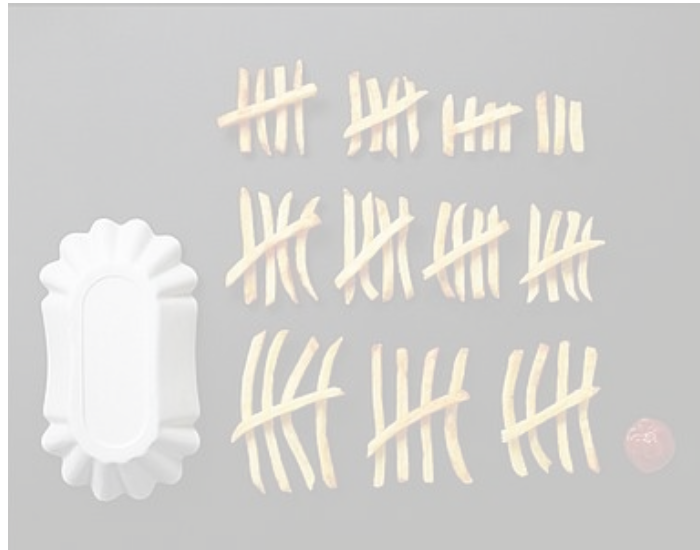
Directories and file naming structure

- Data for each subject are organized in subdirectories labeled "**sub-<label>**"
- Each subject contains subdirectories for different data (type) types
- For a data file that was collected from a given **subject**, the file name **MUST** begin with the string **sub-<label>** and **MUST** contain a task-<label>

```
├─ README.md
├─ dataset_description.json
├─ participants.json
├─ participants.tsv
└─ sub-001
    └─ eeg/
        └─ sub-001_task-GoNoGo.eeg
```

Dataset info

EEG



```
├─ README.md
├─
├─ dataset_description.json
├─ participants.json
├─ participants.tsv
├─ sub-001
│   └─ ses-01
│       ├── eeg/
│       └─ motion/
├─ ses-02
└─ sub-002
```

Modality agnostic data

dataset_description.json

- ├ README.md
- ├ dataset_description.js
- ├ participants.json
- ├ participants.tsv
- └ sub-001
 - ├ ses-01
 - └ ses-02
- └ sub-002

on



```
{  
  "Name": "The mother of all experiments",  
  "BIDSVersion": "1.4.0",  
  "DatasetType": "raw",  
  "License": "CC0",  
  "Authors": [ "Paul Broca", "Carl  
Wernicke" ],  
  "Funding": [ „NIH F37823MFH1" ],  
  "EthicsApprovals": [„HRPO (Protocol  
AR0928" ],  
  "DatasetDOI": "10.0.2.3/dfjj.10"  
}
```

Participants.tsv/.json

- ├ README.md
- ├ dataset_description.json
- ├ **participants.tsv**
- ├ **participants.json**
- └ sub-001
 - ├ ses-01
 - └ ses-02
- └ sub-002

participants.tsv

participant_id	age	group
sub-001	34	C
sub-002	12	P
sub-003	33	M

participants.json

```
"age": {  
  "Description": "age of the  
  participant", "Units":  
  "years"  
},  
"group": {  
  "Description": „assigned group”,  
  "Levels": {  
    „C”: „control”,  
    „M”: „medication”,  
    „P”: „placebo”,  
  }  
}
```

Dataset info

EEG

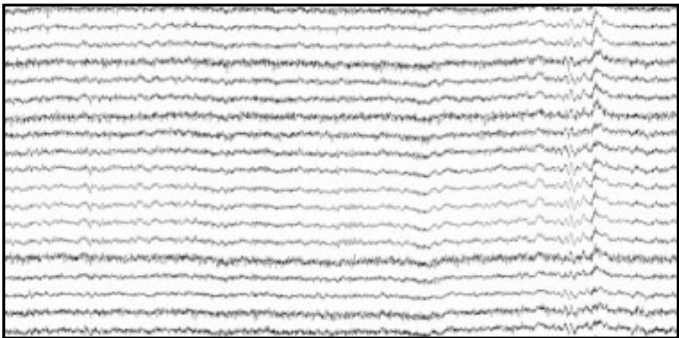


```
├─ README.md
├─
├─ dataset_description.json
├─ participants.json
├─ participants.tsv
├─ sub-001
│   └─ ses-01
│       ├── eeg/
│       └─ motion/
├─ ses-02
└─ sub-002
```

Modality specific data

Overview | eeg

└ sub-001
└ ses-01
└ eeg/



```
{  
  "TaskName": "TASKNAME",  
  "SamplingFrequency": 1000,  
  "SoftwareFilters": "n/a",  
  "EEGChannelCount": 4,  
  "EOGChannelCount": 1,  
  "EEGReference": "placed on Cz",  
  "PowerLineFrequency": 50  
}
```

- └ sub-001_ses-01_task-<label>_eeg.<extension>
- └ sub-001_ses-01_task-<label>_eeg.json
- └ sub-001_ses-01_task-<label>_channels.tsv
- └ sub-001_ses-01_task-<label>_electrodes.tsv
- └ sub-001_ses-01_task-<label>_electrodes.tsv

name	type	units	status	status_description
CP5	EEG	microV	good	n/a
FC5	EEG	microV	bad	high freq noise
FC1	EEG	microV	good	n/a
C3	EEG	microV	good	n/a
VEOG	EOG	microV	good	n/a

name	x	y	z	impedance
CP5	-0.77	-0.30	0.57	8
FC5	-0.77	0.30	0.57	12
FC1	-0.29	0.31	0.91	2
C3	-0.59	0.00	0.81	5
VEOG	n/a	n/a	n/a	n/a

```
{  
  "EEGCoordinateSystem": "T1w",  
  "EEGCoordinateUnits": "mm",  
  "AnatomicalLandmarkCoordinates": {  
    "LPA": [-0.067, 1.736e-09, -3.844e-09],  
    "NAS": [-4.11e-09, 0.091, -4.541e-10],  
    "RPA": [0.064, -6.435e-09, -4.566e-09]  
  },  
  "AnatomicalLandmarkCoordinateSystem": "T1w",  
  "AnatomicalLandmarkCoordinateUnits": "mm",  
  "IntendedFor": "sub-01_T1w.nii.gz"  
}
```

Metadata | eeg.json

REQUIRED

TaskName, SamplingFrequency,
PowerlineFrequency,
EEGReference, SoftwareFilters

RECOMMENDED

TaskDescription, Manufacturer,
EEGChannelCount,
EOGChannelCount, ...

OPTIONAL

-

```
{  
  TaskName: Seeing stuff,  
  SamplingFrequency: 2400,  
  Manufacturer: Brain Products,  
  ManufacturersModelName: BrainAmp DC,  
  EEGChannelCount: 87,  
  EOGChannelCount: 2,  
  TriggerChannelCount: 1,  
  PowerLineFrequency: 50,  
  EEGPlacementScheme: 10-20,  
  EEGReference: single electrode placed on FCz,  
  SoftwareFilters: {  
    Anti-aliasing filter: {  
      half-amplitude cutoff (Hz):  
        500,  
      Roll-off: 6dB/Octave  
    }  
  }  
}
```

Metadata | channels.tsv

REQUIRED

name, type, units

RECOMMENDED

-

OPTIONAL

description, sampling_frequency

name	type	units	description	reference	status	status_description	
VEOG	VEOG	uV	left eye	VEOG-, VEOG+	good	n/a	
FDI	EMG	uV	left first dorsal interosseous	FDI-, FDI+	good	n/a	
Cz	EEG	uV	n/a	REF		bad	high
frequency noise							
UADC001	MISC	n/a	envelope of audio signal	n/a	good	n/a	

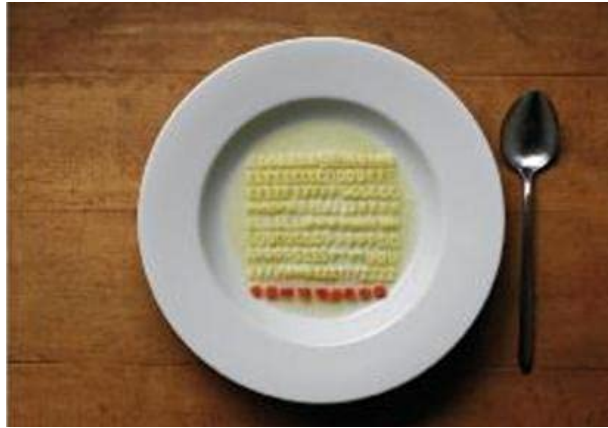
Metadata | electrodes.tsv

name	x	y	z	type	material	color
VEOG+	n/a	n/a	n/a	cup	Ag/AgCl	blue
VEOG-	n/a	n/a	n/a	cup	Ag/AgCl	white
FDI+	n/a	n/a	n/a	cup	Ag/AgCl	red
FDI-	n/a	n/a	n/a	cup	Ag/AgCl	red
GND	-0.07	0.00	-0.070	clip-on	Ag/AgCl	pink
Cz	0.00	0.07	0.06	cup	Ag/AgCl	
	yellow					
REF	-0.07	-0.02	-0.01	cup	Ag/AgCl	grey

Column name	Requirement Level	Data type	Description
name	REQUIRED	string	Name of the electrode contact point. Values in <code>name</code> MUST be unique. This column must appear first in the file.
x	REQUIRED	number	Recorded position along the x-axis. This column must appear second in the file.
y	REQUIRED	number	Recorded position along the y-axis. This column must appear third in the file.
z	REQUIRED	number or "n/a"	Recorded position along the z-axis. This column must appear fourth in the file.
type	RECOMMENDED	string	Type of the electrode (for example, cup, ring, clip-on, wire, needle). This column may appear anywhere in the file.
material	RECOMMENDED	string	Material of the electrode (for example, Tin, Ag/AgCl, Gold). This column may appear anywhere in the file.
impedance	RECOMMENDED	number	Impedance of the electrode, units MUST be in kOhm. This column may appear anywhere in the file.
Additional Columns	OPTIONAL	n/a	Additional columns are allowed if they are defined in the associated metadata file.

Dataset info

Motion



- ├─ README.md
- ├─ dataset_description.json
- ├─ participants.json
- ├─ participants.tsv
- └─ sub-001
 - ├─ ses-01
 - ├─ eeg/
 - └─ **motion/**
 - └─ ses-02
- └─ sub-002

Modality specific data

Files | motion

└─ sub-001/
 └─ ses-01/
 └─ eeg/
 ...
 └─ motion/

└─ sub-001_ses-01_task-<label>_tracksys-<label>_motion.tsv
└─ sub-001_ses-01_task-<label>_tracksys-<label>_motion.json
└─ sub-001_ses-01_task-<label>_channels.tsv

0,2634511	0,092295	0,0086682	0,9305117	0,690106	0,8098815
0,694520	0,1918243	0,8437273	0,3975710	0,885496	0,8952724
0,0766395	0,2587211	0,5434792	0,2822837	0,2789791	0,2326254
0,5779993	0,0456141	0,0490745	0,9408899	0,1533421	0,6683652
0,054556	0,7915927	0,5871733	0,4669577	0,9754468	0,0480541
0,966024	0,1962834	0,7114406	0,3389448	0,7194495	0,4384892
0,984172	0,5079461	0,1180168	0,7966978	0,1753768	0,4886533
0,9883907	0,1557346	0,8002013	0,6334882	0,7526906	0,8529441
...	...	...	...	...	...

```
{
  "SamplingFrequency": 60,
  "SamplingFrequencyEffective": 60.19,
  "TaskName": "BIDS Motion fictive example",
  "TrackingSystemName": "imu1",
  "TaskDescription": "walking and talking",
  "MotionChannelCount": 6,
  "SubjectArtefactDescription": "n/a",
  "TrackedPointsCount": 2,
  "ACCELChannelCount": 3,
  "GYROChannelCount": 3,
  "Manufacturer": "BWSensing",
  "ManufacturersModelName": "BW-imu600",
}
```

name	placement	component	type	tracked_point	units
imu1_rf_acc_x	right_foot		x	ACCEL rf	m/s ²
imu1_rf_acc_y		y	ACCEL	rf	m/s ² right_foot
imu1_rf_acc_z		z	ACCEL	rf	m/s ² right_foot
imu1_rf_gyro_x		x	GYRO	rf	rad/s right_foot
imu1_rf_gyro_y		y	GYRO	rf	rad/s right_foot
imu1_rf_gyro_z		z	GYRO	rf	rad/s right_foot

Metadata | **motion.json**

REQUIRED

TaskName, SamplingFrequency

RECOMMENDED

TaskDescription, RecordingDuration,
MotionChannelCount, <type>ChannelCount,
SoftwareFilters, ...

OPTIONAL

Manufacturer, RecordingSoftware, ...

```
{  
  "TaskName": "BIDS Motion fictive example",  
  "SamplingFrequency": 60,  
  "TrackingSystemName": "IMU Right Hand",  
  "TaskDescription": "walking and talking",  
  "InstitutionAddress": "Fictive address",  
  "InstitutionName": "Fictive Institution",  
  "MotionChannelCount": 18,  
  "RecordingDuration": 4667.641106,  
  "SubjectArtefactDescription": "n/a",  
  "TrackedPointsCount" : 2,  
  "ACCELChannelCount": 6,  
  "GYROChannelCount": 6,  
  "MAGNChannelCount": 6,  
  "Manufacturer": "BWSensing",  
  "ManufacturersModelName": "BW-IMU600",  
}
```

Metadata | channels.tsv

REQUIRED

name, component, type, tracked_point, units

RECOMMENDED

placement, reference_frame

OPTIONAL

status, status_description, sampling_frequency

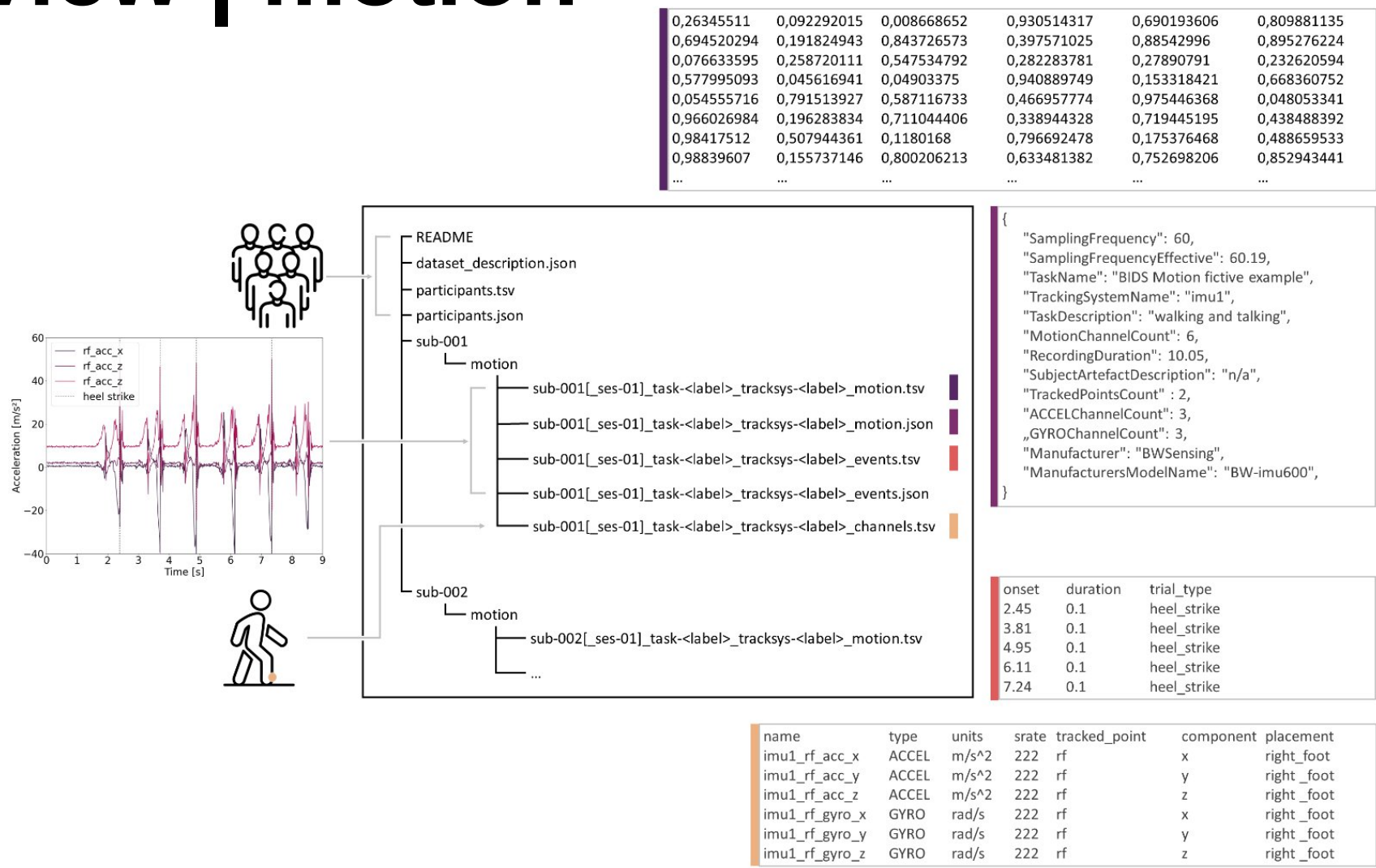
Restricted keyword list for channel type

Restricted keyword list for column `type` in alphabetic order. No

Keyword	Description
ACCEL	Accelerometer channel, one channel for each spatial axis. Collected in the *_channels.tsv file (x, y, or z).
ANGACCEL	Angular acceleration channel, one channel for each spatial axis. Collected in the *_channels.tsv file (x, y, or z).
GYRO	Gyrometer channel, one channel for each spatial axis. Collected in the *_channels.tsv file (x, y, or z).
JNTANG	Joint angle channel between two fixed axis belonging to two different proximal and distal bodypart in deg.

name	component	type	tracked_point	units	reference_frame
t1_acc_x	x	ACCEL	LeftFoot	m/s^2	global
t1_acc_y	y	ACCEL	LeftFoot	m/s^2	global
t1_acc_z	z	ACCEL	LeftFoot	m/s^2	global
t1_gyro_x	x	GYRO	LeftFoot	rad/s	global
t1_gyro_y	y	GYRO	LeftFoot	rad/s	global
t1_gyro_z	z	GYRO	LeftFoot	rad/s	global
...					

Overview | motion



0,26345511	0,092292015	0,008668652	0,930514317	0,690193606	0,809881135
0,694520294	0,191824943	0,843726573	0,397571025	0,88542996	0,895276224
0,076633595	0,258720111	0,547534792	0,282283781	0,27890791	0,232620594
0,577995093	0,045616941	0,04903375	0,940889749	0,153318421	0,668360752
0,054555716	0,791513927	0,587116733	0,466957774	0,975446368	0,048053341
0,966026984	0,196283834	0,711044406	0,338944328	0,719445195	0,438488392
0,98417512	0,507944361	0,1180168	0,796692478	0,175376468	0,488659533
0,98839607	0,155737146	0,800206213	0,633481382	0,752698206	0,852943441
...	...	...	...	...	...

Dataset info

Physio



Physio and some extra stuff

- └─ README.md
- └─ dataset_description.json
- └─ participants.json
- └─ participants.tsv
 - └─ sub-001
 - └─ ses-01
 - └─ eeg/
 - └─ motion/
 - └─ ses-02
 - └─ sub-002

Physio data

└─ sub-01/
 └─ func/
 └─ sub-01_task-nback_physio.json
 └─ sub-01_task-nback_physio.tsv.gz

34	110	0
44	112	0
23	100	1

```
{  
  "Columns": ["cardiac", "respiratory", "trigger"],  
  "Manufacturer": "Brain Research Equipment Ltd.",  
  "SamplingFrequency": 100.0,  
  "StartTime": -22.345,  
  "cardiac": {  
    "Description": "continuous pulse measurement",  
    "Units": "mV"  
  },  
  "respiratory": {  
    "Description": "continuous measurements by respiration belt",  
    "Units": "mV"  
  },  
  "trigger": {  
    "Description": "continuous measurement of the scanner trigger  
signal",  
    "Units": "V"  
  }  
}
```


Task events

sub-<label>/

<data_type>/

<matches>_events.tsv

<matches>_events.json

```
{
  "channel": {
    "Description": "Channel(s) associated with the event"
  },
  "annots": {
    "LongName": "Annotations",
    "Description": "Annotations associated with channels indicated in the channel column.",
    "Levels": {
      "musc": "Muscle artifact. A very common, high frequency, sharp artifact that corresponds with agitation/nervousness in a patient."
    }
  }
}
```

onset	duration		trial_type	response_time	stim_file	channel	annots
1.23	0.65	start	1.435	red_square.jpg	n/a	n/a	
5.65	0.65	stop	1.739	blue_square.jpg	n/a	n/a	
12.1	2.35	n/a	n/a	n/a	Cz	musc	

scans.tsv

sub-<label>/

[ses-<label>/]

sub-<label>[_ses-<label>]_scans.tsv

filename

eeg/sub-control01_task-nback.eeg
eeg/sub-control01_task-walking.eeg
motion/sub-control01_task-walking.tsv

acq_time

1877-06-15T13:55:33 *nback.eeg
1877-06-15T13:55:33 *walking.eeg
1877-06-15T13:56:27 *walking.tsv

Column name	Requirement Level	Data type	Description
filename	REQUIRED	string	Relative paths to files. There MUST be exactly one row for each file. Values in filename MUST be unique. This column must appear first in the file.
acq_time	OPTIONAL	string	Acquisition time refers to when the first data point in each run was acquired. Furthermore, if this header is provided, the acquisition times of all files from the same recording MUST be identical. Datetime format and their anonymization are described in Units. This column may appear anywhere in the file.
Additional Columns	OPTIONAL	n/a	Additional columns are allowed.

BIDS Starterpack

Website

[<https://bids-specification.readthedocs.io/en/stable/>]

Tutorials

[MNE, fieldtrip, EEGLab]

Validator

[<https://bids-standard.github.io/bids-validator/>]

Community

[https://juliuswelzel.github.io/igor_bids_consultation/]

MNE-BIDS 0.16.0 document... News Install Use API CLI Contribute

BIDS V

IGOR BIDS consultations hour Meetings Materials

IGOR BIDS consultations hour

It's a match!

 **BIDS** BRAIN IMAGING DATA STRUCTURE +  **IGOR** Biological Psychology and Neuropsychology

What is the IGOR BIDS Consultation Hour?

We're are happy to offer the IGOR BIDS Consultation Hour, a monthly forum designed to help researchers like you navigate the Brain Imaging Data Structure (BIDS) format. During these sessions, you can connect with experienced BIDS users to get individualized feedback and solutions for your specific BIDS data conversion challenges.

What can I expect?

- **Personalized guidance:** Our experienced researchers, [Julius](#) (EEG & motion data expertise) and [Lennart](#) (MRI expertise), will be available to answer your questions and provide tailored advice on converting your datasets to the BIDS format.
- **Open forum:** Feel free to bring any BIDS-related questions you have, big or small. We're here to help you overcome conversion hurdles and ensure your data adheres to BIDS best practices.
- **Recurring sessions:** The IGOR BIDS Consultation Hour takes place on the 4th Friday of each month at 12:00 PM (noon) CET.

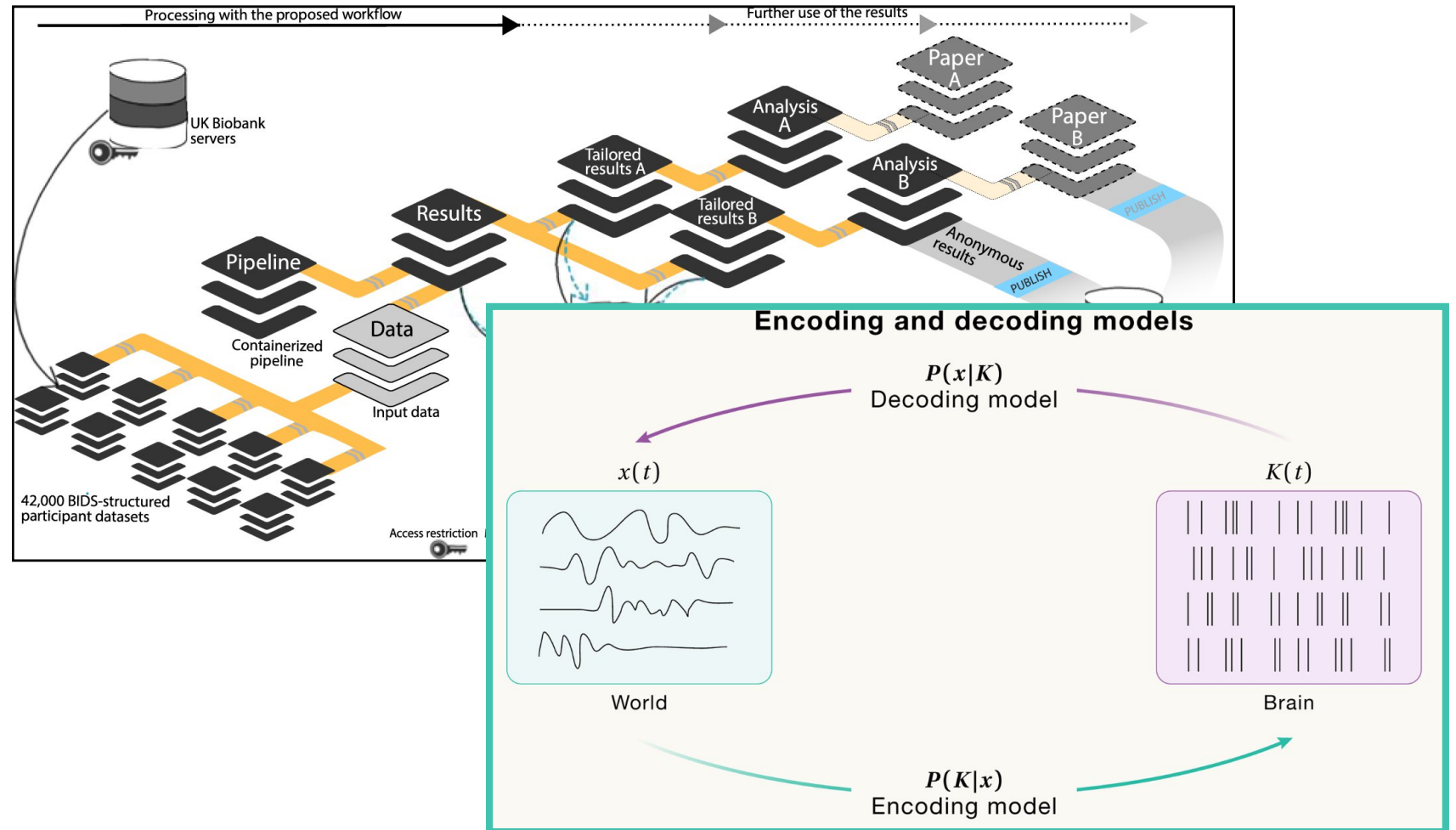
read BIDS datasets Convert MNE sample data to BIDS format interactive data inspection and bad channel selection Convert EEG data to BIDS format

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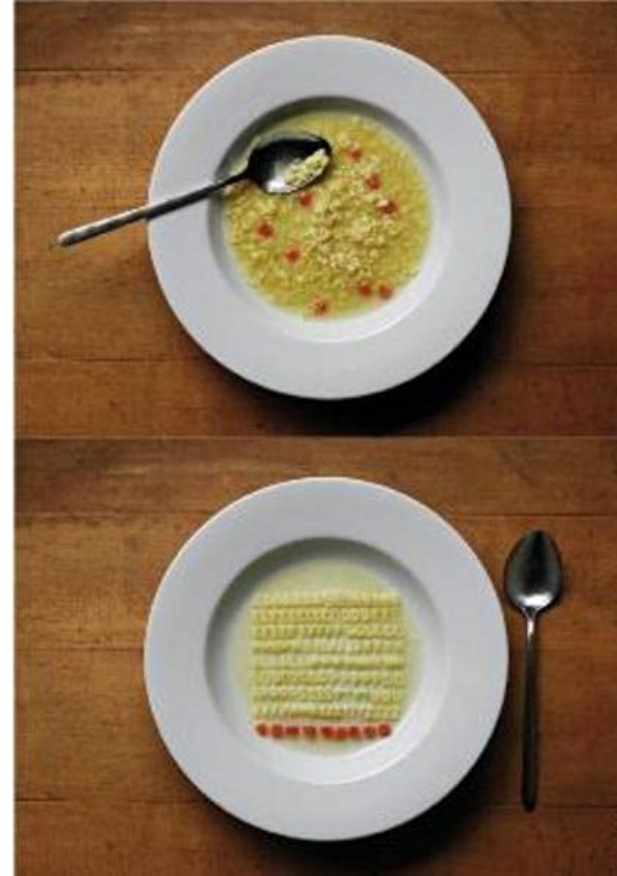
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The
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BIDS in the bigger picture

- Building a sustainable research environment
- Connect rich context information with neural recordings



Conclusions



Thanks



BIDS maintainers

BIDS steering committee

All participants who ever provided data which is now in the BIDS
format



Lara Papin

Hands-On session

https://github.com/JuliusWelzel/BidsWS_TUDresden

How to combine EEG and motion data?

